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Enroll No.....

Class RollNo.....

First Mid Semester Examination,

Data Science Toolkit (CS-601)

Branch: CSE-Semester: VI

Time 1:30 Hrs

Max. Marks 20

Q.1

- a) Define **Data Preprocessing** and explain its role in the data science lifecycle.
b) Explain **Data Discretization** with a real-world example. (CO1 – 5 marks)

OR

- a) A retail organization is struggling with inconsistent and missing data. As a data scientist, how would you apply **Data Cleaning and Transformation** techniques to resolve this issue?
b) Compare **Data Reduction** and **Data Integration** in the context of preprocessing.

Q.2

Evaluate the role of descriptive statistics like mean, standard deviation, skewness, and kurtosis in exploratory data analysis (EDA). How do tools like pivot tables and heat maps enhance insights?

OR

Design a **Data Security Strategy** for a healthcare data science project. Highlight the risks, security issues, and preventive techniques. (CO1 – 5 marks)

- Q.3 a)** List five common roles in a **Data Science** project and describe their responsibilities.
b) Explain the evolution of data science from traditional analytics to AI-driven solutions. (CO2 – 5 marks)

OR

- a) You are part of a data science team building a fraud detection model. What steps would you follow in the **Data Science Lifecycle**?
b) Differentiate between **Exploratory Data Analysis (EDA)** and **Data Modeling** with suitable examples.

Q.4

Assess the impact of **Python libraries** like NumPy, Pandas, and Scikit-learn in building robust data science solutions. Justify their usage with one example each. (CO2 – 5 marks)

OR

Create a case-based scenario where a business applies **Logistic Regression and Ensemble Methods** for customer churn prediction. Highlight the benefits and limitations of each method.